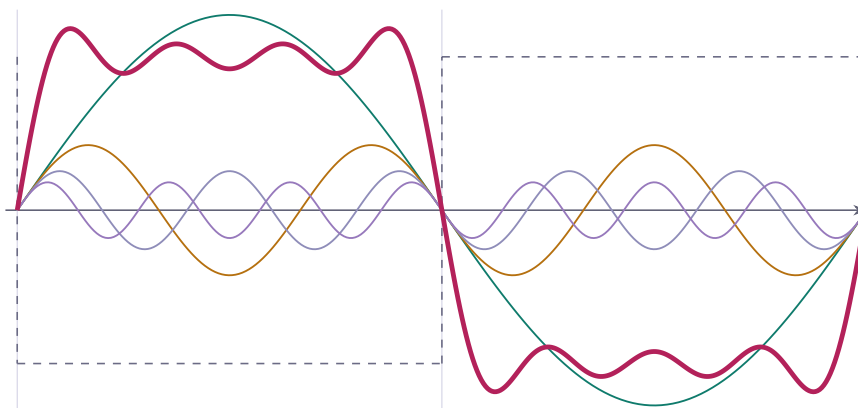


Fourier Series

سری فوریه



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1.1 Periodic Functions

$$f(x + p) = f(x) \quad \Rightarrow \quad f(x + np) = f(x)$$

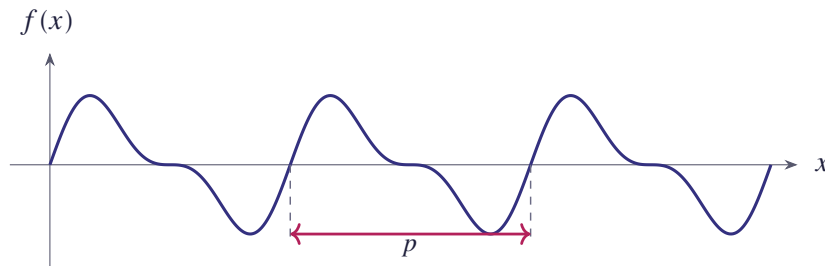


Figure 1.1 A periodic function with period p

- The smallest positive value of p for which the function is periodic is called the fundamental period of the function.
- If a function is periodic with period p , it is also periodic with period np , where n is a natural number.

■ The Periodic Functions Sine and Cosine

Example 1.1

The function $\sin x$ is periodic with period 2π , and $\sin 2x$ is periodic with fundamental period π , although 2π is also a period of it. Likewise $\sin 3x$ is periodic with period $2\pi/3$, although 2π is also a period of it.

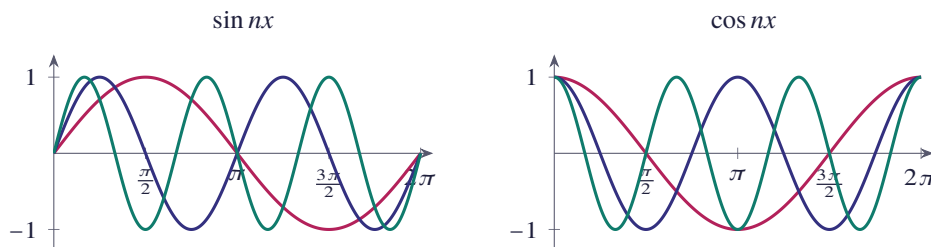


Figure 1.2 The functions $\sin nx$ and $\cos nx$ for $n = 1, 2, 3$ on $[0, 2\pi]$

1.2 Review of Trigonometry

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2} \quad \cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$$

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$

1.3 Review: Integrals and Orthogonality

$$\int_{-\pi}^{\pi} \sin nt \, dt = 0, \quad n \in \mathbb{Z}$$

$$\int_{-\pi}^{\pi} \cos nt \, dt = 0, \quad n \in \mathbb{Z} \setminus \{0\}$$

Proof.

$$\int_{-\pi}^{\pi} \sin nt \, dt = -\frac{1}{n} \cos nt \Big|_{-\pi}^{\pi} = 0, \quad \int_{-\pi}^{\pi} \cos nt \, dt = \frac{1}{n} \sin nt \Big|_{-\pi}^{\pi} = 0 \quad (n \neq 0)$$

Orthogonality

$$\int_{-\pi}^{\pi} \sin nt \sin mt \, dt = 0, \quad n, m \in \mathbb{Z}, n \neq m$$

$$\int_{-\pi}^{\pi} \cos nt \cos mt \, dt = 0, \quad n, m \in \mathbb{Z}, n \neq m$$

$$\int_{-\pi}^{\pi} \sin nt \cos mt \, dt = 0, \quad n, m \in \mathbb{Z} \quad (n \neq m \text{ or } n = m)$$

Proof.

$$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(n-m)x \, dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos(n+m)x \, dx$$

$$\int_{-\pi}^{\pi} \cos nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos(n-m)x \, dx$$

$$\int_{-\pi}^{\pi} \sin nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin(n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \sin(n-m)x \, dx$$

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt = \pi, \quad \int_{-\pi}^{\pi} \cos^2 nt \, dt = \pi, \quad n \in \mathbb{Z} \setminus \{0\}$$

Proof.

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt = \int_{-\pi}^{\pi} \frac{1 - \cos 2nt}{2} \, dt = \pi, \quad \int_{-\pi}^{\pi} \cos^2 nt \, dt = \int_{-\pi}^{\pi} \frac{1 + \cos 2nt}{2} \, dt = \pi$$

■

1.4 The Fourier Series

Remark ■ History

Joseph Fourier (1768–1830) put forward the idea of expressing periodic functions as a series of sinusoidal functions. When this revolutionary idea was presented, some mathematicians, Lagrange among them, objected to its generality, until Dirichlet stated the conditions for the convergence of the Fourier series.

Theorem 1.2 ■ The Fourier series

If the function $f(x)$ is periodic with period 2π , it can be expressed as a trigonometric series as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

Proof.

Suppose the periodic function has been written as the trigonometric series above; it suffices to find the coefficients a_0 , a_n and b_n .

Computing a_0 . Integrating both sides over an interval of length 2π gives

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) dx &= \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] dx \\ &= a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left(\underbrace{a_n \int_{-\pi}^{\pi} \cos nx dx}_0 + \underbrace{b_n \int_{-\pi}^{\pi} \sin nx dx}_0 \right) = 2\pi a_0 \\ \Rightarrow a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx\end{aligned}$$

a_0 is the mean value or DC level of the function.

Computing a_n . Multiplying both sides by $\cos mx$ and integrating over an interval of length 2π :

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) \cos mx dx &= \int_{-\pi}^{\pi} a_0 \cos mx dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} a_n \cos nx \cos mx dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} b_n \sin nx \cos mx dx = \pi a_m \\ \Rightarrow a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx\end{aligned}$$

Computing b_n . Multiplying both sides by $\sin mx$ and integrating over an interval of length 2π :

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) \sin mx dx &= \int_{-\pi}^{\pi} a_0 \sin mx dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} a_n \cos nx \sin mx dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} b_n \sin nx \sin mx dx = \pi b_m \\ \Rightarrow b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx\end{aligned}$$

■

Key result 1.3 ■ The Fourier series and Euler's formulas

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{2\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{2\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{2\pi} f(x) \sin nx dx$$

(the integral is taken over one period of length 2π .)

Remark

At points of discontinuity of the function, the value of the Fourier series equals the average of the upper and lower limits at that point.

1.5 Review: The Definite Integral of a Product of Two Functions

By integration by parts:

$$\int_{\alpha}^{\beta} f(x)g(x) dx = \left[f(x)G_1(x) - f^{(1)}(x)G_2(x) + f^{(2)}(x)G_3(x) - + \dots \right]_{\alpha}^{\beta}$$

where $f^{(k)}$ is the k -th derivative of f and G_k is the k -th integral of g :

derivatives of f		integrals of g
$f(x)$	$\searrow$	$g(x)$
$f^{(1)}(x)$	$\searrow$	$G_1(x)$
$f^{(2)}(x)$	$\searrow$	$G_2(x)$
$f^{(3)}(x)$	$\searrow$	$G_3(x)$
$\vdots$		$\vdots$

1.6 Examples of Fourier Series

Example 1.4

Find the Fourier series of the following function:

$$f(x) = \begin{cases} -k & -\pi < x < 0 \\ k & 0 < x < \pi \end{cases} \quad f(x+2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos nx dx + \int_0^{\pi} k \cos nx dx \right] \\ &= \frac{1}{\pi} \left[-k \frac{\sin nx}{n} \Big|_{-\pi}^0 + k \frac{\sin nx}{n} \Big|_0^{\pi} \right] = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin nx dx + \int_0^{\pi} k \sin nx dx \right] \\ &= \frac{1}{\pi} \left[k \frac{\cos nx}{n} \Big|_{-\pi}^0 - k \frac{\cos nx}{n} \Big|_0^{\pi} \right] = \frac{k}{n\pi} [\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi} (1 - \cos n\pi) \end{aligned}$$

$$\cos n\pi = \begin{cases} -1 & n \text{ odd} \\ 1 & n \text{ even} \end{cases} \quad \Rightarrow \quad 1 - \cos n\pi = \begin{cases} 2 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

$$b_1 = \frac{4k}{\pi}, \quad b_2 = 0, \quad b_3 = \frac{4k}{3\pi}, \quad b_4 = 0, \quad b_5 = \frac{4k}{5\pi}, \dots$$

$$f(x) = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$$

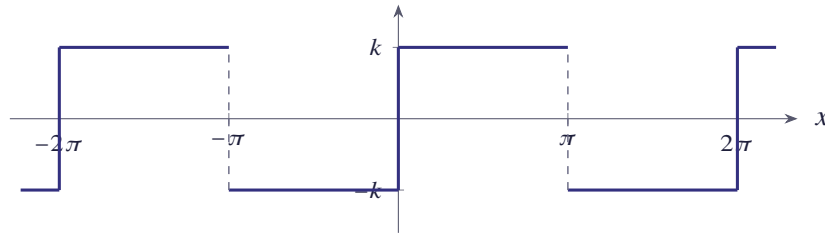


Figure 1.3 A square wave $f(x)$ with period 2π and values $\pm k$

The partial sums of the series:

$$S_1 = \frac{4k}{\pi} \sin x, \quad S_2 = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x \right), \quad S_3 = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x \right)$$

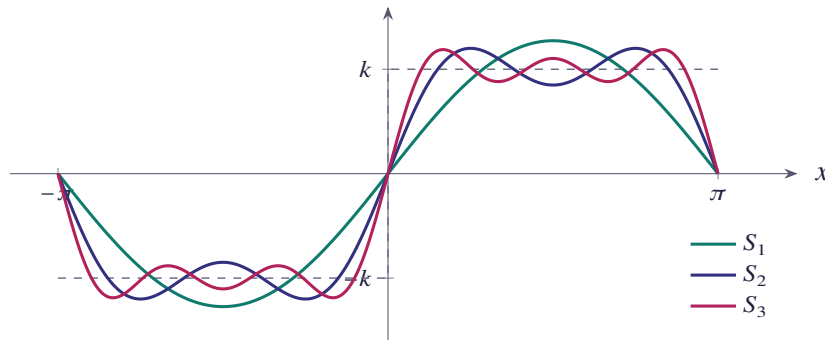


Figure 1.4 The partial sums S_1 , S_2 and S_3 of the Fourier series of the square wave (the dashed line is the square wave itself)

Setting $x = \pi/2$:

$$f(\pi/2) = \frac{4k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right) = k \quad \Rightarrow \quad \pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

Example 1.5

Find the Fourier series of the following function:

$$f(x) = x, \quad -\pi \leq x \leq \pi, \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx dx = \frac{1}{\pi} \left[x \left(\frac{1}{n} \sin nx \right) - (1) \left(-\frac{1}{n^2} \cos nx \right) \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[\frac{1}{n^2} (\cos n\pi - \cos(-n\pi)) \right] = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx dx = \frac{2}{\pi} \int_0^{\pi} x \sin nx dx = \frac{2}{\pi} \left[x \left(-\frac{1}{n} \cos nx \right) - (1) \left(-\frac{1}{n^2} \sin nx \right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[-\frac{\pi}{n} \cos n\pi \right] = \frac{2}{n} (-1)^{n+1} \end{aligned}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin nx = 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right)$$

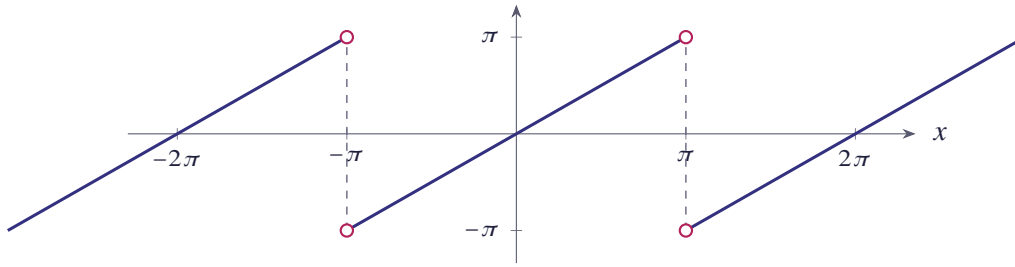


Figure 1.5 The sawtooth wave $f(x) = x$ on $[-\pi, \pi]$ with its periodic extension

Setting $x = \pi/2$:

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \frac{\pi}{2} = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin\left(n\frac{\pi}{2}\right) = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right) = \sum_{n=1}^{\infty} \frac{2}{2n-1} (-1)^{n+1} \Rightarrow \pi \\ &= \sum_{n=1}^{\infty} \frac{4}{2n-1} (-1)^{n+1} \end{aligned}$$

The value of the series at the point of discontinuity $x = \pi$:

$$f(\pi) = ? \Rightarrow \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi) = 0$$

Key result 1.6 ■ Important remarks

1. If a periodic function is odd, all the a coefficients are zero.
2. At a point of discontinuity, the value of the Fourier series equals the average of the upper and lower limits at that point.

Example 1.7

Find the Fourier series of the following function:

$$f(x) = x^2, \quad -\pi \leq x \leq \pi, \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{2\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{1}{3} x^3 \right]_0^{\pi} = \frac{\pi^2}{3}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx \\ &= \frac{2}{\pi} \left[x^2 \left(\frac{1}{n} \sin nx \right) - 2x \left(\frac{-1}{n^2} \cos nx \right) + 2 \left(\frac{-1}{n^3} \sin nx \right) \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{2\pi}{n^2} \cos n\pi \right] = \frac{4}{n^2} (-1)^n \end{aligned}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = 0$$

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$$

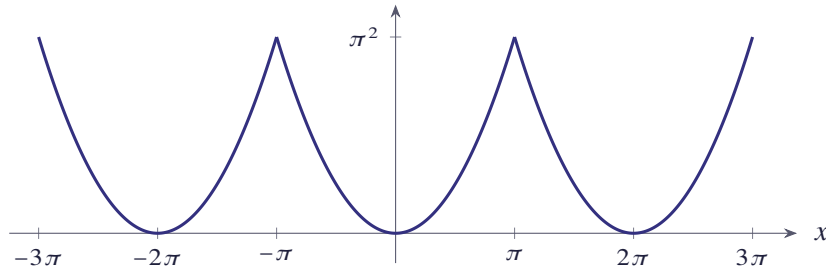


Figure 1.6 The function $f(x) = x^2$ on $[-\pi, \pi]$ with its periodic extension

Key result 1.8 ■ Important remark

If a periodic function is even, all the b coefficients are zero.

Setting $x = 0$:

$$f(0) = 0 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos(n \cdot 0) \quad \Rightarrow \quad \frac{\pi^2}{3} = \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^{n+1}$$

$$\pi^2 = \sum_{n=1}^{\infty} \frac{12}{n^2} (-1)^{n+1} = 12 \left(1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots \right)$$

Example 1.9

Find the Fourier series of the following function:

$$f(x) = x^2, \quad 0 \leq x \leq 2\pi, \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} x^2 dx = \frac{1}{2\pi} \left[\frac{1}{3} x^3 \right]_0^{2\pi} = \frac{1}{2\pi} \cdot \frac{1}{3} 8\pi^3 = \frac{4\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx dx = \frac{1}{\pi} \left[x^2 \left(\frac{1}{n} \sin nx \right) - 2x \left(\frac{-1}{n^2} \cos nx \right) + 2 \left(\frac{-1}{n^3} \sin nx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} (2 \times 2\pi) \left(\frac{1}{n^2} \cos 2\pi n \right) = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx = \frac{1}{\pi} \left[x^2 \left(\frac{-1}{n} \cos nx \right) - 2x \left(\frac{-1}{n^2} \sin nx \right) + 2 \left(\frac{1}{n^3} \cos nx \right) \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[(4\pi^2) \left(\frac{-1}{n} \right) + 2 \cdot \frac{1}{n^3} - 2 \cdot \frac{1}{n^3} \right] = -\frac{4\pi}{n}$$

$$f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

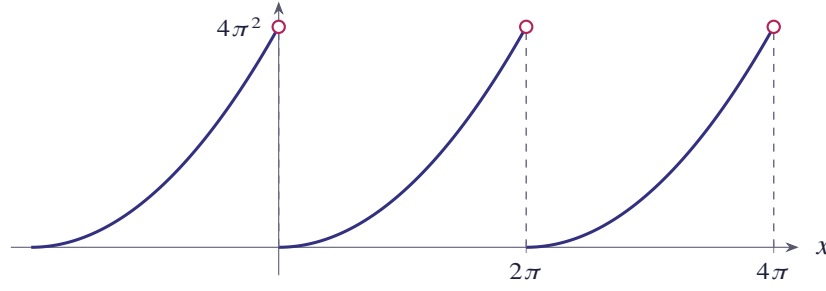


Figure 1.7 The function $f(x) = x^2$ on $[0, 2\pi]$ with its periodic extension

1.7 Fourier Series of Functions with an Arbitrary Period

$$f(x + T) = f(x)$$

Using the auxiliary variable $t = \frac{2\pi}{T}x$ ($x : 0 \rightarrow T$ and $t : 0 \rightarrow 2\pi$):

$$f\left(\frac{T}{2\pi}t\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt$$

$$a_0 = \frac{1}{2\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) dt, \quad a_n = \frac{1}{\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) \cos nt dt, \quad b_n = \frac{1}{\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) \sin nt dt$$

Returning to $x = \frac{T}{2\pi}t$:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T}nx + b_n \sin \frac{2\pi}{T}nx$$

$$a_0 = \frac{1}{2\pi} \int_T f(x) \frac{2\pi}{T} dx = \frac{1}{T} \int_T f(x) dx$$

$$a_n = \frac{1}{\pi} \int_T f(x) \cos\left(\frac{2\pi}{T}nx\right) \frac{2\pi}{T} dx = \frac{2}{T} \int_T f(x) \cos\left(\frac{2\pi}{T}nx\right) dx$$

$$b_n = \frac{1}{\pi} \int_T f(x) \sin\left(\frac{2\pi}{T}nx\right) \frac{2\pi}{T} dx = \frac{2}{T} \int_T f(x) \sin\left(\frac{2\pi}{T}nx\right) dx$$

Theorem 1.10 ■ The Fourier series (period T)

If the function $f(x)$ is periodic with period T , it can be expressed as a trigonometric series as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T}nx + b_n \sin \frac{2\pi}{T}nx, \quad f(x + T) = f(x)$$

$$a_0 = \frac{1}{T} \int_T f(x) dx, \quad a_n = \frac{2}{T} \int_T f(x) \cos \frac{2\pi}{T}nx dx, \quad b_n = \frac{2}{T} \int_T f(x) \sin \frac{2\pi}{T}nx dx$$

Remark

At points of discontinuity of the function, the value of the Fourier series equals the average of the upper and lower limits there; the value of the Fourier series at the point x is

$$\frac{f(x^+) + f(x^-)}{2}$$

1.8 Dirichlet's Theorem**Theorem 1.11 ■ Dirichlet's theorem**

If the function $f(x)$ is periodic and **bounded**, has a **finite number of maxima and minima**, and also a **finite number of points of discontinuity** in each period, then this function has a Fourier series, and the value of the Fourier series at points of discontinuity equals the average of the right and left limits at those points.

Examples of functions that do not satisfy Dirichlet's conditions:

$$f(x) = 1/x, \quad 0 < x \leq 1, \quad f(x+1) = f(x)$$

$$f(x) = \sin(1/x), \quad 0 < x \leq 1, \quad f(x+1) = f(x)$$

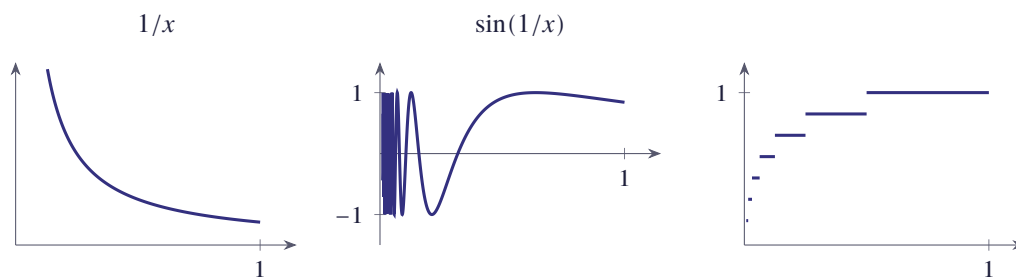


Figure 1.8 Functions that violate Dirichlet's conditions: $1/x$ (unbounded), $\sin(1/x)$ (infinitely many maxima and minima) and a staircase function with infinitely many steps

1.9 Solving Some Problems

Example 1.12 ■ Problem 1

Find the Fourier series of the following function:

$$f(x) = \sin \alpha x, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Since this function is odd, $a_0 = 0$ and $a_n = 0$.

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin \alpha x \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin \alpha x \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\cos(\alpha - n)x - \cos(\alpha + n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(\alpha - n)x}{\alpha - n} - \frac{\sin(\alpha + n)x}{\alpha + n} \right]_0^{\pi} = \dots = \frac{2n(-1)^n \sin \alpha \pi}{\pi(\alpha^2 - n^2)} \end{aligned}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2n(-1)^n \sin \alpha \pi}{\pi(\alpha^2 - n^2)} \sin nx$$

We examine the series when $\alpha = 1$:

$$\lim_{\alpha \rightarrow 1} b_1 = \lim_{\alpha \rightarrow 1} \frac{2(-1)^1 \sin \alpha \pi}{\pi(\alpha^2 - 1^2)} = \lim_{\alpha \rightarrow 1} \frac{-2(\pi \cos \alpha \pi)}{\pi(2\alpha)} = 1, \quad b_n|_{n \neq 1} = 0$$

Hence the Fourier series is simply $f(x) = \sin x$.

Exercise 1.13

Examine the series when $\alpha = m$, $m \in \mathbb{N}$.

Example 1.14 ■ Problem 2

Find the Fourier series of the following function and deduce $\csc \alpha \pi$ from it.

$$f(x) = \cos \alpha x, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Since this function is even, $b_n = 0$ and $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$.

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \alpha x \, dx = \frac{1}{\pi} \int_0^{\pi} \cos \alpha x \, dx = \frac{1}{\pi} \left[\frac{\sin \alpha x}{\alpha} \right]_0^{\pi} = \frac{1}{\alpha \pi} \sin \alpha \pi$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} \cos \alpha x \cos nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\cos(\alpha - n)x + \cos(\alpha + n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(\alpha - n)x}{\alpha - n} + \frac{\sin(\alpha + n)x}{\alpha + n} \right]_0^{\pi} = \dots = \frac{(-1)^n 2\alpha \sin \alpha \pi}{\pi(\alpha^2 - n^2)} \end{aligned}$$

$$f(x) = \frac{1}{\alpha \pi} \sin \alpha \pi + \frac{2\alpha \sin \alpha \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \cos nx$$

The periodic function is continuous at $x = 0$, so

$$\begin{aligned} f(0) = 1 &= \frac{1}{\alpha \pi} \sin \alpha \pi + \frac{2\alpha \sin \alpha \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \quad \Rightarrow \quad \frac{1}{\sin \alpha \pi} = \csc \alpha \pi \\ &= \frac{1}{\alpha \pi} + \frac{2\alpha}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \end{aligned}$$

Exercise 1.15

Prove that if $\alpha = m$, $m \in \mathbb{N}$, then

$$a_0 = 0, \quad a_n = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}$$

Example 1.16 ■ Problem 3

Find the Fourier series of the following function and deduce $\coth \alpha \pi$ from it.

$$f(x) = e^{-\alpha x}, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Lemma 1:

$$\int e^{\alpha x} \cos nx \, dx = \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \cos nx + n \sin nx)$$

Proof: integrating by parts twice, with $I = \int e^{\alpha x} \cos nx \, dx$:

$$\begin{aligned} I &= \frac{1}{\alpha} e^{\alpha x} \cos nx - \int \frac{1}{\alpha} e^{\alpha x} (-n \sin nx) \, dx = \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha} \int e^{\alpha x} \sin nx \, dx \\ &= \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha} \left[\frac{1}{\alpha} e^{\alpha x} \sin nx - \int \frac{1}{\alpha} e^{\alpha x} (n \cos nx) \, dx \right] \\ &= \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha^2} e^{\alpha x} \sin nx - \frac{n^2}{\alpha^2} I \\ \Rightarrow \quad I &= \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \cos nx + n \sin nx) \end{aligned}$$

Lemma 2:

$$\int e^{\alpha x} \sin nx \, dx = \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \sin nx - n \cos nx)$$

(proof left as an exercise.)

The function is neither even nor odd, so all the Fourier coefficients must be computed.

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \, dx = \frac{1}{-2\pi\alpha} [e^{-\alpha x}]_{-\pi}^{\pi} = \frac{1}{2\pi\alpha} [e^{\alpha\pi} - e^{-\alpha\pi}] = \frac{\sinh \alpha\pi}{\alpha\pi} \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \cos nx \, dx = \left\{ \frac{e^{-\alpha x}}{\pi(\alpha^2 + n^2)} (-\alpha \cos nx + n \sin nx) \right\}_{-\pi}^{\pi} \\ &= \frac{\alpha(-1)^n}{\pi(\alpha^2 + n^2)} [e^{\alpha\pi} - e^{-\alpha\pi}] = \frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \sin nx \, dx = \left\{ \frac{e^{-\alpha x}}{\pi(\alpha^2 + n^2)} (-\alpha \sin nx - n \cos nx) \right\}_{-\pi}^{\pi} \\ &= \frac{n(-1)^n}{\pi(\alpha^2 + n^2)} [e^{\alpha\pi} - e^{-\alpha\pi}] = \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \\ f(x) &= \frac{\sinh \alpha\pi}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \cos nx + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \sin nx \right) \\ &= \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \cos nx + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sin nx \right) \right) \end{aligned}$$

Since $x = \pi$ is a point of discontinuity, the value of the Fourier series there equals the average of the upper and lower limits of the function at that point, i.e.

$$\frac{f(\pi^+) + f(\pi^-)}{2} = \frac{e^{\alpha\pi} + e^{-\alpha\pi}}{2} = \cosh \alpha\pi$$

$$\cosh \alpha\pi = \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \cos n\pi + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sin n\pi \right) \right)$$

$$= \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \frac{2\alpha}{\pi(\alpha^2 + n^2)} \right)$$

$$\coth \alpha\pi = \frac{\cosh \alpha\pi}{\sinh \alpha\pi} = \frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \frac{2\alpha}{\pi(\alpha^2 + n^2)}$$